

Milk Hill 2001 - Measurement Basis and Scale Derivation

Controlling physical measurements and equations used by the completed 409-circle reconstruction.

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1. Controlling measurements

Quantity	Control
Overall formation diameter	787 ft
Formation radius	$787 / 2 = 393.5$ ft
Largest circle diameter	72 ft
Smallest circle diameter	$40 \text{ in} = 40 / 12 = 3.333333\dots$ ft
Total circles	409
Organization	1 center + 6 arms x 68 circles = 409

2. Scale equations

Formation radius: $R = D / 2 = 787 / 2 = 393.5$ ft.

Minimum diameter conversion: $d_{\min} = 40 \text{ in} / 12 = 3.333333\dots$ ft.

Circle count identity: $N = 1 + 6 \times 68 = 409$.

Mean conversion from normalized image radius to feet in the completed register: $s = \text{mean}(r_{\text{ft}} / r_{\text{norm}}) = 165.429639119$ ft per normalized-radius unit.

Ground coordinate conversion used by the completed register: $x_{\text{ft}} = r_{\text{ft}} \cos(\theta)$, $y_{\text{ft}} = r_{\text{ft}} \sin(\theta)$, with the sign convention inherited from the rectified image coordinate system.

Outer-envelope check: $R_{\text{edge}} = \max(r_{\text{center}} + d/2)$.

3. Independent physical checks

Check	Calculated	Required	Residual / status
Outermost edge radius	393.49995 ft	393.50000 ft	-0.00005 ft
Minimum diameter	$3.3333 \text{ ft} / 40.00 \text{ in}$	40.00 in	PASS
Maximum diameter	72.0000 ft	72.00 ft	PASS
Circle count	409	409	PASS
Arm population	68 each	68 each	PASS

Conclusion

The completed physical register is internally consistent with the three controlling measurements. The outer envelope closes on the 393.5-ft radius to within 0.00005 ft, while the diameter range reaches the 40-in minimum and 72-ft maximum exactly at the stated precision.